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Detection of *E. coli* IncX1 Plasmid-Mediated *mcr-5.1* Gene in an Indian Hospital Sewage Water Using Shotgun Metagenomic Sequencing: A First Report

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Colistin is used against a multitude of multidrug-resistant and extremely drug-resistant Gram-negative bacterial infections. The emergence of colistin resistance is highly concerning as it may lead to the failure of this last-resort antibiotic. Since the identification of first mobile colistin resistance (*mcr*) genes, several variants of *mcr* genes have been reported, but still there are limited studies detecting *mcr* genes in hospital sewage water. The prevalence of *mcr* in the hospital environment is extremely hazardous putting health care workers, patients, and visitors at a higher risk of exposure. It may lead to a multidrug-resistant bacterial infection outbreak. In this study, we report *mcr-5.1* gene in an Indian hospital sewage water using shotgun metagenomics, as a first report. The *mcr-5.1* gene in the metagenome has been explored using RGI, ABRicate, NCBI database, CARD, and Resfinder. This *mcr-5.1* gene harbored by *Escherichia coli* is a plasmid-mediated gene carried by an IncX1 plasmid pSGMCR103. The bioinformatics analysis revealed the genetic environment of *mcr-5.1* gene, which consisted of mobile element protein, ChrB domain protein, putative major facilitator superfamily type transporter, and a hypothetical protein.

Keywords: colistin, *mcr* gene, shotgun sequencing, hospital sewage, metagenome

Introduction

Colistin or polymyxin E is the last-resort antibiotic against multidrug-resistant and extremely drug-resistant Gram-negative bacterial infections.[1] The last decade has witnessed a rapid surge in the emergence of plasmid-mediated, mobile colistin-resistant mobile colistin resistance (*mcr*) genes in Enterobacterales. Since the first detection of *mcr-1* gene in *Escherichia coli* and *Klebsiella pneumoniae* in animals and humans in China, 10 major *mcr* gene determinants (*mcr-1* to *mcr-10*) and several *mcr* gene variants have been detected in environmental samples besides animals and humans.[2,3] The *mcr* genes have been reported in urban wastewater, hospital sewage water, and food-producing animals, as well as their derived food products like beef, chicken, and pork.[4–6]

Four variants of *mcr-5* have been detected to date (Fig. 1). The *mcr-5* is of 1,644 bp size and was detected in *Salmonella Paratyphi* B dTa+ isolates for the first time.[7] These isolates were randomly selected from the bacterial samples of poultry and food submitted at German National Reference Laboratory for the Analysis and Testing of Zoonoses (NRL Salmonella). It was identified as a part of 7,337 bp transposon of the Tn3 family, usually located on multicopy ColE-type plasmids. It have been reported in *E. coli* isolates from diseased pigs in Japan[8] and from animals[9] and the human vaginal microbiome in China.[10] The *mcr-5.2* has been identified in *E. coli* isolated from intestinal tract of pigs.[11]

The variation in *mcr-5.2* was conferred by a deletion of three nucleotides encoding an amino acid in the central part of the protein, whereas *mcr-5.3* was reported in a CTX-M-8-producing *E. coli* isolate (ECPB39) from a diseased horse

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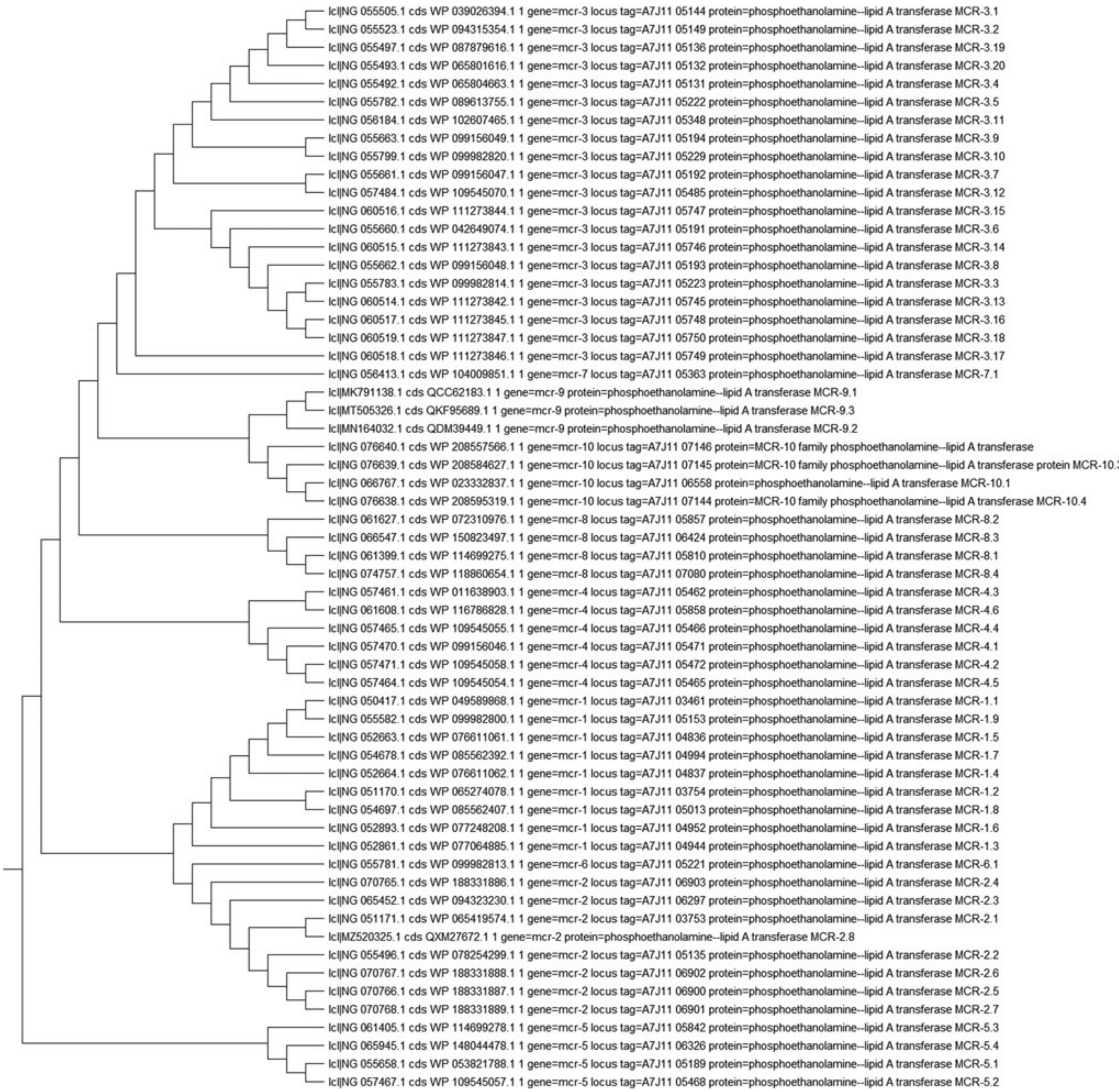

FIG. 1. Phylogenetic tree of sequences of all variants of *mcr* genes obtained from the NCBI. The evolutionary history was inferred using the UPGMA method. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used for inferring the phylogenetic tree. mcr, mobile colistin resistance.

that succumbed to pneumonia.[12] It was mediated by 5,361 bp plasmid (pECPB39, belonging to an unknown replicon type). It emerged due to Ala414Ser mutation, while *mcr-5.4* gene has been reported in a hospital water environment with the most probable host identified as Pseudomonadales.[13]

Materials and Methods

Sample collection from hospital sewage water

To investigate various antimicrobial resistance markers in hospital settings, six sewage water samples were collected from six different hospitals located in various regions of India during the time period of December 2018 to March 2021. Two liters of nonprocessed, unfiltered sewage water was collected in sterile bottles from different spots of the main hospital sewage pipeline, where all the sewage effluents of different sites of hospital were drained. The sample was placed in an icebox with ice packs and in a sterile environment, the sample was vortexed. Around 50 mL was collected in a sterile bottle and transported with ice packs to the laboratory for sequencing. Water samples were centrifuged at 7,000 *g* in 15 mL falcon to separate cell pellet and water. The pellet was stored at −20°C.

DNA extraction and shotgun metagenome sequencing

The DNA was extracted using DNeasy PowerSoil Kit (Catalog No. 12888-100; Qiagen) by following manufacturer's protocol. The DNA was finally eluted in 20 μL of Solution C6 of the kit. The DNA samples were checked for quality, using NanoDrop™ 2000 Spectrophotometer (ND2000; Thermo Fisher Scientific), to confirm that DNA is free of contaminants like RNA and proteins. Furthermore, to check the integrity, 50 ng of DNA along with Lambda DNA/HindIII Marker (SM0102; Thermo Fisher Scientific) was loaded on a 1% agarose gel and stained with UltraPure™ Ethidium Bromide (10 mg/mL concentration, Catalog No. 15585011; Thermo Fisher Scientific) followed by electrophoresis by running the gel at 80V for 1 hour. Finally, the gel was visualized in a SmartView Pro 1100 Imager System (UVCI-1100; Major Science).

DNA concentrations were determined by using Qubit™ dsDNA HS Assay Kit (Catalog No. Q32853; Thermo Fisher Scientific), which contains dye as well as buffer and two standards. Qubit reagent contains one of the most sensitive dyes, which accurately quantify only the double-stranded DNA molecules. Dye and the buffer were diluted at 1:200 ratio and 199 μL of the working solution was transferred into a Qubit Assay Tube (Catalog No. Q32856; Thermo Fisher Scientific) to which 1 μL of the DNA sample was mixed, vortexed, and incubated at RT for 2 minutes. The final volume in each tube would be 200 μL. The readings were taken in a Qubit 3.0 Fluorometer (Thermo Fisher Scientific). The DNA quality was found to be intact with 154.8 ng DNA yield. The extracted DNA was subjected to shotgun metagenome sequencing using Illumina HiSeq, 2×150 bp paired-end run.[14]

Data quality check and contig assembly

Data quality was checked using FastQC and MultiQC software. The trimmed reads were processed using Kraken2 and assembled using Megahit v1.1.3 with—k-min 31—k-max 141 and—k-step 28 parameters.[15–18] Contigs shorter than 200 bp were removed from the assembly so that the contigs have properly paired reads with at least 50 bp alignment. The assembly quality was checked by bowtie2.[19]

Annotation of the assembled contigs and identifying taxonomic distribution

The assembled contigs were checked for homology against NCBI nucleotide database. Taxonomic distribution of BLAST hit contigs was identified with TaxonKit v0.2.3.[20] The eukaryotic contigs were removed and remaining were annotated using Prokka v1.14.6 with—metagenome option.[21]

Identification of AMR markers

To detect the AMR markers, the Resistance Gene Identifier (RGI) v5.2.0 was run under the ''Perfect'' parameter to get the perfect matches with the AMR sequences in the Comprehensive Antibiotic Resistance Database (CARD) (https://github.com/arpcard/rgi).[22] For further confirmation, ABRicate v1.0.1 was run using the predownloaded databases (https://github.com/tseemann/abricate).[23] ABRicate was run using CARD database, NCBI, and Resfinder database.

Analysis of the genetic environment of mcr-5.1 *gene*

The nucleotide BLAST search was performed on the whole contig (>k141_458493) of length 4131 bp to reveal the genetic environment. PlasFlow (https://github.com/smaegol/PlasFlow) and PlasClass (https://github.com/Shamir-Lab/PlasClass) were run to know if the gene was of chromosomal or plasmid origin.

Results

Out of the six collected Indian hospital sewage water samples, the DNA collected from sixth sample, Sample_6 metagenome, was found to be harboring the *mcr-5.1* gene. This sample was collected from Domkal Super Specialty and Sub divisional Hospital, Domkal, Murshidabad, West Bengal, on March 21, 2021, at 5.30 am Indian Standard Time (IST).

The RGI predicted the presence of *mcr-5.1* gene in the Sample_6 metagenome with 100% identity and 100% coverage. The *mcr-5.1* gene was further confirmed by running command-line tool ABRicate, which screens massive number of metagenome contigs for AMR as well as virulence genes. ABRicate against CARD database predicted the contig >k141_458493 to be encoding *mcr-5* gene (Accession ID KY807921.1:4941–6549) with gene length ranging from 1,521 to 3,128 bp, coverage length 1–1,608/1,608, 100% coverage, and 100% identity.

This was further predicted to be phosphoethanolamine-lipid A transferase *mcr-5.1* (Accession ID A7J11_05189) against the NCBI database, with the corresponding alignment region ranging from 1,521 to 3,164 bp, coverage length 1–1,644/1,644, 100% identity, and 100% coverage. ABRicate using Resfinder predicted the same result, that is, alignment of 1–1,644/1,644 bp against *mcr*-5.1 gene (Accession ID KY80792) with 100% identity and 100% coverage.[24] The nucleotide BLAST search was performed on the whole contig (>k141_458493) of length 4,131 bp listing *E. coli* strain ENV103 plasmid pSGMCR103 (MK731977.1) as the best result with 99% identity, 0% expect value, and 0% gaps.

The contig aligned from 188 to 4,131 bp with 9,780 bp to 13,723 bp of the plasmid pSGMCR103 (Fig. 2A, C). The plasmid backbone was composed of 58,124 bp (Fig. 2B). The orientation of alignment was negative (−). The length of 1–187 bp remain unaligned. The plasmid pSGMCR103 is an IncX1 replicon type mostly found in Enterobacterales, present in a narrow host range.[4] This makes *mcr-5.1* distinct from other *mcr* genes carried by *E. coli*, which are often mediated by IncX4 plasmids.[4] The genetic environment of *mcr-5.1* gene reported in this study consisted of a mobile element protein, a ChrB domain protein that confers resistance to chromate, a putative major facilitator superfamily-type transporter, and a hypothetical protein (Fig. 2B, D).

Due to the lack of information about the presence of Tn3-like transposable elements from our contig, the transmissibility of this gene could not be inferred. Nonetheless, in the light of earlier studies and the presence of a truncated mobile element protein, the transferability of this gene cannot be ruled out. In addition, PlasFlow predicted the contig to be a plasmid belonging to Proteobacteria and PlasClass predicted it to be a plasmid with 0.818 score.

The assembled contigs were checked for homology against NCBI nucleotide database. Taxonomic distribution

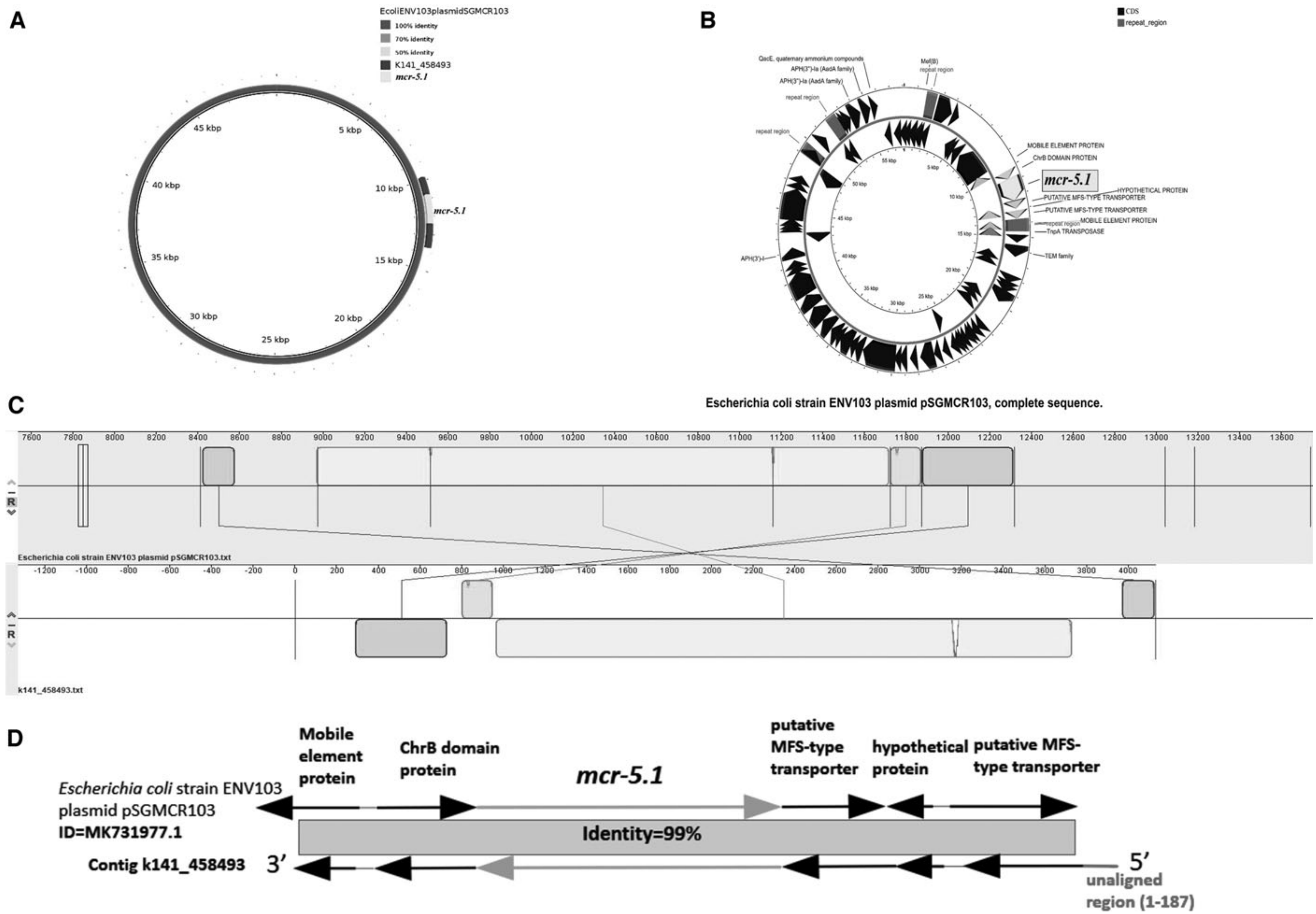

FIG. 2. The genetic analysis of contig >k141_458493 with the detected *mcr-5.1* gene. **(A)** The alignment of contig >k141_458493 (188 to 4,131 bp) with *Escherichia coli* strain ENV103 plasmid pSGMCR103 (9,780 to 13,723 bp) (drawn usinghttp://brig.sourceforge.net/). **(B)** *E. coli* strain ENV103 plasmid pSGMCR103 with *mcr-5.1* gene (http://cgview.ca/). **(C)** The DNA alignment of contig >k141_458493 with *E. coli* strain ENV103 plasmid pSGMCR103 (http://darlinglab.org/mauve/mauve.html). **(D)** The genetic environment of the detected *mcr-5.1* gene in contig >k141_458493 (https://inkscape.org/)

of BLAST hit contigs was identified with TaxonKit v0.2.3.[20] The most probable host for *mcr-5.1* gene was predicted to be *E. coli* (Phylum Proteobacteria, Class Gammaproteobacteria, Order Enterobacterales, and family Enterobacteriaceae).

Discussion

To the best of our knowledge, this is the first report of *mcr-5.1* gene detection in a hospital sewage water sample from India, which is an alarming signal for emergence of new variants for colistin resistance. The *mcr-5* was first identified in d-tartrate fermenting *Salmonella enterica subsp. enterica* serovar *Paratyphi B* isolated from poultry and food products.[7] It was reported to be transposon associated with only 34–36% amino acid identity common with *mcr-1, mcr-2, mcr-3,* and *mcr-4* (Fig. 1).[7] Earlier, the variant *mcr-5.1* has been detected in ready-to-eat chicken rice in Singapore and water sources from the Western Cape, South Africca.[4,5]

The *mcr-1, mcr-3,* and *mcr-5* genes have been detected in environmental water sources, but to date, only *mcr-5.4* gene has been reported from hospital wastewater.[13] The *mcr*-5.1 identified is carried by an *E. coli* plasmid as predicted by the BLAST search against NCBI nucleotide database, PlasFlow and PlasClass. The plasmid-mediated colistin resistance genes have higher transmissibility due to horizontal gene transfer (HGT). The HGT aids the dissemination of *mcr-5.1* in the environment. The detection of the *mcr-5.1* gene thriving in hospital settings is highly concerning. It aggravates the probability of colistin-resistant bacterial infections among health care personnel working there, patients, as well as visitors.[25] The Domkal Super Speciality and Sub divisional Hospital, Domkal, is a public hospital where people from nearby areas come for treatment. This hospital with high number of patients can serve as the breeding ground for novel AMR markers and highly virulent strains of MDR bacteria.

If we look into the demography of Murshidabad, with a low literacy rate of 63.88% (as per Census 2011) and 80.22% rural population, it is a poverty-stricken area with majority of residents belonging to the labor class.[26] It becomes significant to consider the socioeconomic condition of an area in context to the AMR profile in the environment of that region. Several studies have shown that all these factors are often correlated and elevate the risk of infections with multidrug-resistant organisms.[27] The combination of several factors like poverty, illiteracy, high patient to hospital ratio, lack of hygiene in hospitals, self-purchasing of drugs, and sale of WHO AWaRe group of antibiotics without legitimate prescription may create a condition highly conducive for the spread of AMR genes.

The lack of closed drainage system and inefficient waste water treatment strategies in this area can spread the AMR to drinking water as well, risking lives leading to a potential MDR bacterial infection outbreak. Surveillance of AMR in an environment is necessary for adopting a policy to tackle it as well as to update the existing treatment regimen. In a country like India where ''out-of-counter'' drug selling is a normal practice, it becomes essential to cut down the general use of antibiotics whose resistance is prevalent in the environment, especially the last-resort antibiotics.

Conclusions

The predicted mortality rate of 10 million per year by the year 2050 due to antimicrobial resistance is startling and it becomes more minacious for a country like India, which is considered one of the hotspots for AMR.[28] The hospital sewage water of a region is a mirror reflection of the antibiotic resistance genes and the antibiotic-resistant bacteria circulating in its environment. According to the World Health Organization, robust measures at the hospital level are crucial to curb the AMR dissemination in the environment. The first-time detection of colistin resistance marker *mcr-5.1* in an Indian hospital sewage water through the shotgun metagenome sequencing calls for attention and requires further studies in the hospital and clinical settings. A well-planned global surveillance network to detect antibiotic-resistant bacteria employing high-end next-generation sequencing technologies will play a pivotal role in determining the future strategies to eliminate the probability of a multidrug-resistant bacterial outbreak.

Sequence data

The metagenomic sequences have been submitted on NCBI under the BioProject accession number PRJNA682952.

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Disclosure Statement

No competing financial interests exist.

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